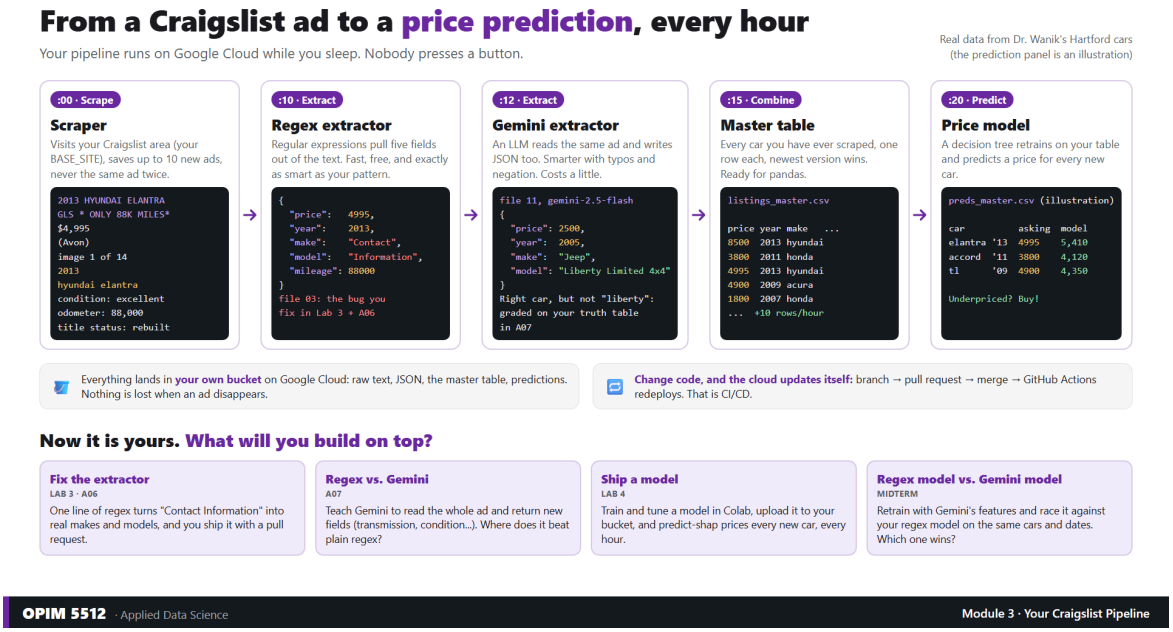
Lab 4: Deploying Your Model to Production · START HERE

> Hartford: **Wed Oct 14** · Stamford: **Wed Oct 21** · Submit by **Thu Oct 22, 11:59 PM** (both campuses).



Solo · about 2 hours · Colab · Dr. Wanik's cleaned Hartford cars, then your own Google Cloud bucket. Your pipeline scrapes, extracts and combines cars every hour. Tonight you **fit a really good model** (tuned, and better than the baseline) and **deploy it** into that pipeline, so it prices every new car on its own, every hour (and saves why, with SHAP).

	What	Link
🚀	**The lab notebook** (open in Colab, then File → Save a copy in Drive)	Lab4_Model_In_Production.ipynb
🗺️	**Tonight in 20 steps** (print it, or keep it on a second screen)	OPIM5512_Lab4_20_Steps_PRINT.pdf
☁️	**Final GCP Guide F26** (if your pipeline is not running yet)	FINAL_GCP_Guide_F26.ipynb
🧠	**The production function** your model runs inside	`cloud_function/predict-shap/main.py`
📊	**The training data** (453 Hartford cars, Sep 28 to Oct 13, after the A06 regex fix)	lab4_listings_master_snapshot.csv

Before class (10 minutes, please)

Get `predict-shap` deploying before you arrive: it takes about 3 minutes. Open your myscrapers repo on GitHub and find your case (Part 0 of the notebook has every step and command):

What you see on your repo's Code tab	Do this
"This branch is N commits **behind** drdave-teaching/myscrapers"	**A:** Sync fork → **Update branch** (never "Discard commits")
"forked from drdave-teaching/myscrapers", 0 behind, and a `cloud_function/predict-shap` folder	**B:** Actions → **Deploy predict-shap** → **Run workflow**
No "forked from drdave-teaching/myscrapers" under the repo name	**C:** paste the Cloud Shell block from Part 0

- [ ] **Deploy predict-shap** shows a green check (A or B), or the Cloud Shell block finished (C)
- [ ] Your **A06 regex fix** is shipped (otherwise half your makes say `contact`)
- [ ] You know your **project ID** and **bucket name** (usually the same)

- [ ] **2-step verification is on** for your Google Cloud Gmail. Since Oct 13, Google Cloud may show a "Turn on 2-step verification" box and can lock you out of the console without it: click **Enable MFA** (or turn it on later at myaccount.google.com → Security), not just "Remind me later"

Tonight in 8 parts

Part	Minutes	You do
0. Before you start	5	Get `predict-shap` deploying (case A, B or C)
1. Load	10	Dr. Wanik's cleaned cars, same versions as production (scikit-learn 1.6.1, shap 0.48.0)
2. Clean	10	`Ford` vs `ford`, `chevy` vs `chevrolet`, flag junk rows (odd price, placeholder mileage)
3. Baseline	10	The numbers to beat: guess the median, and the decision tree your pipeline already trains
4. Tune	25	Tune a decision tree, a random forest and KNN with cross-validation. **You pick the champion**
Your turn	15	**You code it:** actual-vs-predicted scatter, error histogram and summary stats, champion vs baseline
5. Explain	20	SHAP on your champion, then **price a car you know** and see why
6. Ship	20	Save the model + model card, upload to your bucket (your account is checked first), run `predict-shap` right from Colab
7. Your model at work	5	What your frozen model priced in the cloud (open the two CSVs yourself), and why one price
8. Wrap up	5	Five short answers, submit

Where everything lives in your bucket

```
models/model.joblib, model_card.json     your LIVE model (the newest upload; predict-shap reads only this)
models/history/model_<time>.joblib       a copy of every model you ever shipped, named by time (never overwritten)
structured/datasets/listings_master.csv  all your cars (the scraper adds ~10 an hour)
structured/preds_lab4/preds_master.csv   every price your model made, tagged with the model version that made it
structured/preds_lab4/shap_master.csv    the SHAP values behind each of those prices
structured/preds/                        the pipeline's background decision tree (train-dt): ignore it for Lab 4
```

Your model is frozen: production never retrains it. Retraining = run the notebook again and upload; production switches to the new model by itself, and the old predictions stay, tagged with the old version.
Submit: HuskyCT → *Submit Lab 4* (upload your `.ipynb`), due **Thu Oct 22, 11:59 PM** (Hartford and Stamford).
Leave `predict-shap` running. Every hour at :25 it predicts and explains your newest cars, and your midterm can chart how your model's errors and explanations move over time.

Stuck?

Ask Claude: paste the cell, the **full** error, and what you expected. Most problems are one of these:

- **Wrong Google account** in the Colab pop-up: pick the Gmail you used for Google Cloud. Part 6c prints "Signed in as ..." and stops with a clear message if that account cannot see your bucket.
- **`status: no_model`** from `predict-shap`: your upload in Part 6c did not run (check `PROJECT_ID` and `BUCKET_NAME` in Part 1b).
- **Part 7 says your model priced exactly 500 cars**: you have more than 500 in your bucket. Production prices at most 500 per run (40 for KNN); the rest get priced at the next hourly :25 runs. Training is not limited: it uses every car you load.
- **Part 7 says "No production predictions yet"**: run Part 6d (it starts `predict-shap` from Colab and waits for it), then re-run the cell.